

NSF Grant Proposal

Directorate for Biological Sciences

Proposal Number: BIO-2024-99735

Submission Date: November 1, 2024

Project Information

Project Title: Microplastic Transport Dynamics in Pacific Northwest Watersheds

Principal Investigator: Dr. Maria Lopez

Institution: University of Idaho

Department: Department of Environmental Science

Requested Amount: \$320,000

Project Period: January 1, 2026 - December 31, 2028

Sponsoring Agency: National Science Foundation

Research Area: Environmental Science

Project Abstract

This project examines the transport, accumulation, and ecological impacts of microplastic pollution in freshwater systems of the Pacific Northwest. Using a combination of field sampling, laboratory analysis, and hydrological modeling, we will characterize microplastic concentrations across 15 watersheds in Idaho and Washington. The study will identify key transport pathways, seasonal variation in microplastic loading, and relationships between land use and contamination levels. Results will inform regional water quality management strategies and contribute to national databases on freshwater microplastic pollution.